CS 113

Week 8



Agenda

Announcements

Switch Statements

Nested Conditionals

2D Arrays

Go over quiz 1

Announcements

HW #4 - due to Oct 26 at midnight

- Reminder: homework can be submitted up to 2 days after the deadline for 10% penalty each day.
- Failure to submit your work in this timeframe will result in a zero. No exceptions.

Lab today: Tic Tac Toe!

Quiz #2 - on Thurs Oct 29 (Topics TBD)

- You may bring a cheat sheet
- Practice quiz with solutions will be posted to the class website by Thurs Oct 22

Grade Breakdown

Final Exam: 30%

Quizzes: 35%

Homework: 15%

Labs: 15%

Attendance / Participation: 5%

Switch statement: days of the week

```
class Week8 {
  public static void main(String[] args) {
    int day = 3;
    switch (day) {
      case 1:
        System.out.println(x: "Monday");
        break;
      case 2:
        System.out.println(x: "Tuesday");
        break;
      case 3:
        System.out.println(x: "Wednesday");
        break;
      case 4:
        System.out.println(x: "Thursday");
        break;
      case 5:
        System.out.println(x: "Friday");
        break;
      default:
        System.out.println(x: "Weekend or invalid day");
    }
  }
}
```

Switch statement

What if we remove `break` from `case 3`?

```
class Week8 {
  public static void main(String[] args) {
    int day = 3;
    switch (day) {
      case 1:
        System.out.println(x: "Monday");
        break;
      case 2:
        System.out.println(x: "Tuesday");
        break;
      case 3:
        System.out.println(x: "Wednesday");
        // break;
      case 4:
        System.out.println(x: "Thursday");
        break;
      case 5:
        System.out.println(x: "Friday");
        break;
      default:
        System.out.println(x: "Weekend or invalid day");
    }
  }
}
```

Nested Switch Statements

Ex: Order food

First: patron chooses breakfast or lunch

Second: patron chooses dish

Conditionals: Movie tickets

```
class MovieTickets {
  public static void main(String[] args) {
    int age = 17;

    if (age < 18) {
      System.out.println(x: "Youth ticket: $8");
    } else {
      System.out.println(x: "Regular ticket: $15");
    }
  }
}
```

Nested Conditionals: Movie tickets

```
class MovieTickets {
  public static void main(String[] args) {
    int age = 17;
    boolean hasStudentID = true;

    if (age < 18) {
      System.out.println(x: "Youth ticket: $8");
    } else {
      if (hasStudentID) {
        System.out.println(x: "Student ticket: $10");
      } else {
        System.out.println(x: "Regular ticket: $15");
      }
    }
  }
}
```

Nested Conditionals: Movie tickets

```
class MovieTickets {
  public static void main(String[] args) {
    int age = 17;
    boolean hasStudentID = true;

    if (age < 18) {
      System.out.println(x: "Youth ticket: $8");
    } else {

      if (hasStudentID) {
        System.out.println(x: "Student ticket: $10");
        System.out.println(x: "Regular ticket: $15");
      }
    }
  }
}
```

What if age is 16?

What if age is 20 and hasStudentID is false?

Nested Conditionals

The **inner** if is only evaluated if the **outer** else executes.

Indentation makes the structure easier to follow.

Nested conditionals let us make decisions that depend on earlier decisions.

Recall: 1D Arrays

```
int[] nums = {1, 2, 3};
```

Why might we need two dimensions?

Real-world data often lives in grids such as chessboards, tic-tac-toe boards, pixel maps in images, or spreadsheet tables

Visualize the grid: rows and columns

Say we have 3 students, and each student has taken 3 quizzes. We can structure this as a grid (3x3), where each element is a student's quiz grade.

	Student A	Student B	Student C
Quiz 0	70	40	90
Quiz 1	65	66	80
Quiz 2	85	70	95

2D Array: Declaration & Initialization

2D Array Access

2D Array: Modifying Elements

Examples